Chemical Engineering Journal 530 (2026) 173719

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

A theoretical multiscale approach for topology-based toughness design of network polymers

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ARTICLE INFO

Keywords:
Network polymers
Topology modulation
Fracture toughness
Multiscale simulation

ABSTRACT

Network-structured polymers enable high rigidity but suffer from premature fracture due to strain localization at densely crosslinked sites. This work establishes a multiscale framework to quantify toughness enhancement achieved through topological modification of epoxy thermosets using homogeneous and flexible toughening agent. Thermoset microstructures modeled using all-atom molecular dynamics simulations exhibited component-dependent asymmetric damage under tensile loading: hardener segments underwent early intramolecular distortion and scission, whereas the flexible toughener retained extensibility and delayed the covalent-bond rupture. Such component-specific behavior promotes more homogeneous pore nucleation within the network, preserving load-transfer pathways at large strains. Accordingly, epoxy networks with appropriately designed topologies exhibited fracture energies up to 20 times greater than neat epoxy while retaining network homogeneity. Such molecular-level properties were coupled to a finite element model to characterize the influence of continuum-level stress concentrations and crack initiation. It was found that the dissipation of externally applied strain energy is primarily determined by crack-tip blunting and substantial expansion of the plastic zone, thus identifying topological modulation of the microstructure as a key factor in enhancing toughness. The proposed theorical multiscale framework provides a mechanistic link from chain-level rearrangements to macroscopic crack resistance and offers quantitative design guidance for next-generation thermoset polymers that balance stiffness with effective toughness.

1. Introduction

Network-structured polymers, characterized by high crosslinking density, are indispensable for load-bearing applications due to their substantial mechanical rigidity [1,2]. The primary factor contributing to high stiffness is the efficient intermolecular load transfer across the dense crosslinking points, which enhances structural integrity [3–5]. However, this advantage is mitigated by a significant drawback—localized strain concentration within the polymer chains [6], which accelerates fracture initiation and substantially diminishes mechanical toughness. This inherent brittleness renders network network-structured polymers highly susceptible to catastrophic fracture without any discernible precursors under operational conditions.

In response to these limitations, recent research has increasingly focused on optimizing the topology of network structures. Previous investigations have shown that modifying network topology can significantly improve key intrinsic properties, such as flexural strength, yielding stress, failure strain, and Fracture energy [7–10]. A promising strategy to enhance toughness involves the deliberate incorporation of flexible organic components, known as homogeneous toughners, into rigid polymer networks [11–14]. This strategic modification aims to increase the inter-crosslinking distances, thereby mitigating strain concentration at critical points and promoting more efficient energy dissipation under stress. In contrast to nano-compositing approaches that alter the rheological characteristics or heterogeneous toughening agents that promote microphase separation [15,16], homogeneous toughners preserve the inherent processability while maintaining network-level compositional uniformity. This approach, therefore, effectively mitigates the growth of localized cracks without sacrificing the intrinsic nature of the polymer network.

The fracture behavior of network-structured polymers is inherently a multiscale phenomenon. Microscopic damage initiated by chain

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https://doi.org/10.1016/j.cej.2026.173719
Received 31 October 2025; Received in revised form 13 January 2026; Accepted 1 February 2026
Available online 2 February 2026
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distortion and molecular bond failure ultimately evolves into macroscopic crack formation–propagation. To design network topologies that effectively enhance toughness, it is imperative to establish a precise and quantifiable correlation between molecular-level damage mechanisms and macroscopic fracture behavior. However, this integrated perspective presents significant challenges in experimental validation, as the connection between these disparate scales is difficult to capture through conventional techniques. As a result, the critical insights required to optimize network structures for a balanced enhancement of both stiffness and toughness remain underexplored, thereby limiting the potential for the development of advanced materials with superior mechanical performance.

Motivated by the above, this study investigates the fracture resistance mechanics of topologically modified network polymers using a computational multiscale framework. Epoxy microstructures functionalized with a flexible toughening agent were prepared in an all-atomistic molecular dynamics simulation environment. Asymmetric chain damage mechanism between hardener and toughening agent under tensile deformation has been elucidated at the atomic scale. The evolution and coalescence of nanosized voids at the ends of fractured bonds has been explored in detail. Molecular-level damage energy was transferred to the finite element method, which provided a numerical understanding of stress concentration and resulting failure characteristics of the material in the continuum field. Overall, this study provides theoretical guidance for intelligent topological design of network-structured polymer to enhance their effective toughness.

2. Experimental demonstration

To clarify the effect of topological modification on the fracture toughness of the polymers, we first performed experimental demonstrations. The mechanical behavior of epoxy thermoset modified with homogeneous additives was characterized. Bisphenol A diglycidyl ether (DGEBA) and ethylenediamine (EDA) were used as epoxy resin and curing agents to form a three-dimensional network (Fig. 1(a), left). Bis (2-hydroxyethyl)octane-1,8-diyldicarbamate (C-8 diol), synthesized via an aminolysis reaction, was used as a toughening agent [12]. During the curing process, both the hardener and the toughening agent participate in a ring-opening reaction with the epoxide group (Fig. 1(a) middle). Due to its much longer end-to-end length than EDA, C-8 diol acts as an extended crosslinker, bridging resin chains and widening inter-crosslink spacing.

The epoxy specimens were fabricated considering different toughening agent molar ratios (Fig. 1(a) right). Details of the synthesis process for each specimen are provided in Experimental Section of Supplementary Data. The neat samples were prepared with a stoichiometric ratio of epoxide to amine of 2:1 (**Table S1**). Other reinforced samples additionally contain C-8 diol as a toughening agent. Even when toughening agents were added, the stoichiometric ratio of epoxide to its counterpart was maintained at 2:1. In other words, potential effects arising from stoichiometric shifts were effectively eliminated from consideration. The toughening agents in epoxy network were considered to be up to 40 mol% of the total epoxide groups.

The mechanical response of the prepared epoxy specimens was then evaluated through uniaxial tensile tests (**Figs. 1(b)** and **1(c)**). Details of the test procedure were provided in Measurement Section of Supplementary Data. The stress-strain profiles of specimens exhibited the typical behavior of thermoset polymers, i.e., an elastic response without strain softening. Notably, the addition of a C-8 diol toughening agent effectively retarded the fracture point. The fracture strain showed a

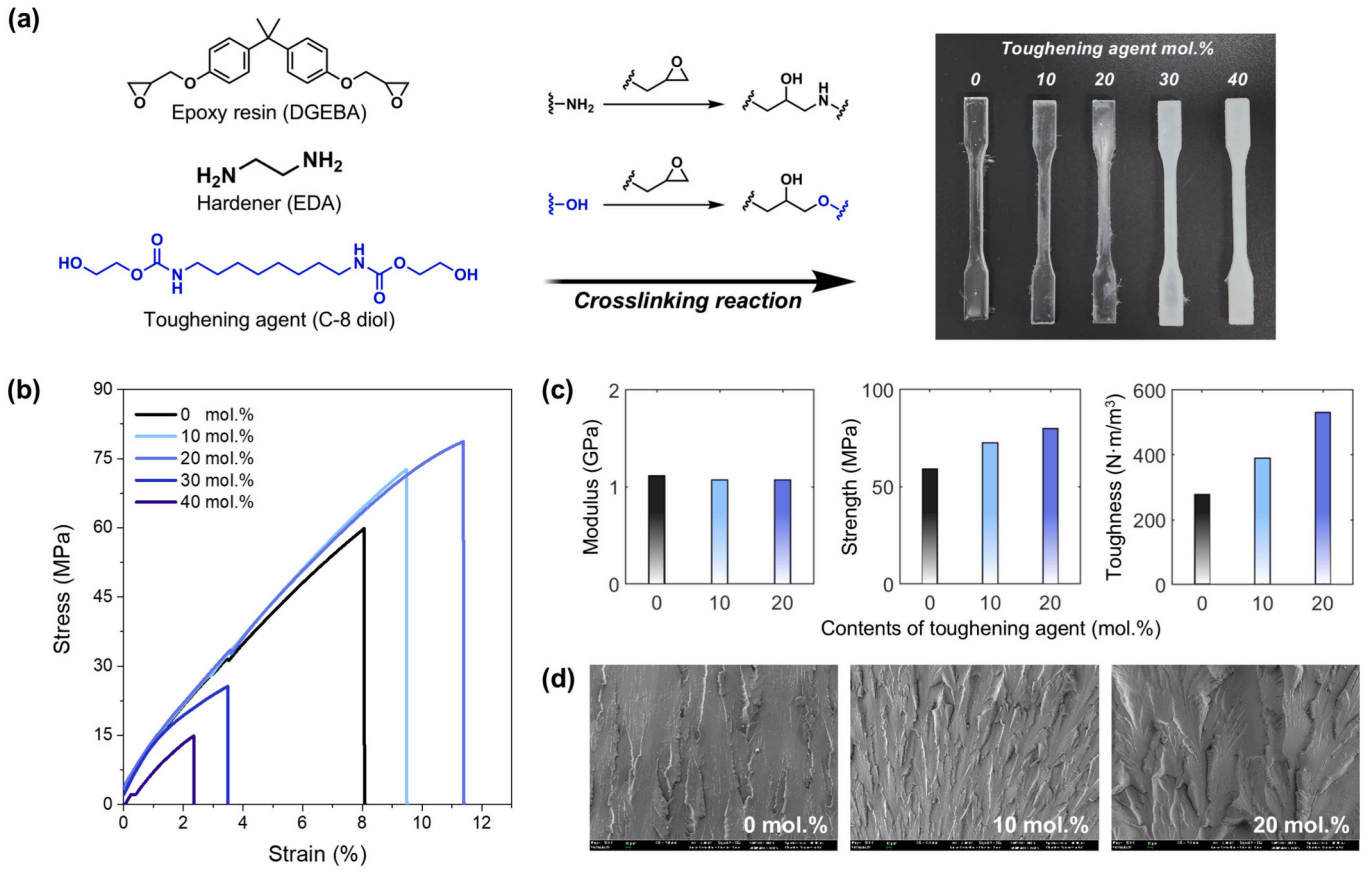


Fig. 1. Toughness enhancement of crosslinked thermosets using network topology modification. (a) Molecular structure of the thermoset component (left) and specimens prepared with various concentrations of toughening agent (right). (b) Stress-strain profiles obtained from uniaxial tensile tests and (c) changes in elastic modulus, fracture strength, and mechanical toughness. (d) Fracture surface morphology of each specimen captured by scanning electron microscope.

1.41-fold increase up to 20 mol% of toughening agents (8.1% for 0 mol% and 11.4% for 20 mol%). Since the slopes of the profiles were preserved at all strain levels, the resulting toughness is significantly improved by a factor of 2.01 (277.71 N·m/m^3 for 0 mol% and 530.84 N·m/m^3 for 20 mol%). When the toughening agent was excessive (over 30 mol%), the mechanical properties deteriorated as the epoxy network fragmented.

For all cases considered, the stress-strain profiles did not show a distinct yield point. In other words, while most of the area of the specimen underwent primarily elastic deformation, localized crack—initiated at structural imperfections such as topological defects, nanovoids, and dangling ends—determined the macroscopic toughness of the material. Therefore, the mechanical behavior obtained thus far explicitly suggested that the toughening agent contributes to retarding the growth and propagation of localized cracks. This was further supported by the fracture surface morphology obtained by scanning electron microscope (SEM) (Fig. 1(d)). As the content of the toughening agent increases, the fracture surface of the specimen becomes rougher (see **Fig. S1** for surface and cross-sectional morphology before and after loading in 0–40 mol% diol additive). The rough cross-section demonstrated that the mechanical stress is not concentrated at the crack tip but rather is distributed over a wide area.

The influence of diol additive on mechanical properties was further identified through loading-unloading tests. The analysis focused on samples containing ~20 mol% diol, corresponding to the composition that exhibited optimal toughening performance. Under cyclic loading, all epoxy systems exhibited progressive structural degradation manifested as a gradual reduction in the stress–strain response (Fig. 2(a)). Notably, when the evolution of peak stress at $\varepsilon = 0.7\%$ was examined, the decay in peak stress was significantly suppressed in the presence of the diol additive (Fig. 2(b)). This behavior indicates that the incorporation of diol units enhances the resistance to cumulative damage under repeated loading. Furthermore, stress relaxation experiments conducted under strain-holding conditions revealed a pronounced acceleration of stress dissipation in diol-modified networks (Fig. 2(c)). In other words, the overall structural resilience of the network is improved through additive-based topology design.

Other analyses performed on the prepared specimens are as follows. The structure of the synthesized C-8 diol was identified by nuclear magnetic resonance (NMR) (**Figs. S2** and **S3**). Differential scanning calorimetry (DSC) analysis confirmed that adding reinforcing agent made the network less rigid and lowered the glass transition temperature (**Fig. S4**). The crosslinking density of the thermoset decreases as the toughening agent increases the distance between the crosslinking sites, which could be obtained by quantifying the crosslink density from dynamic mechanical analysis (DMA) (**Fig. S5**).

From chemical bond cleavage of chains to cavitation and stress concentration at the macroscale, the crack propagation is governed by a multiscale factor. However, experimental approaches alone have limited spatiotemporal resolution, hindering a detailed understanding of the fracture resistance mechanisms associated with network topology modification. Therefore, in the following section, we clarify the strengthening mechanisms of topology-modified networks through a theorical simulation approach. The reactive molecular dynamics and finite element methods are integrated to capture the multiscale fracture process.

3. Multiscale interpretation of toughness enhancement

3.1. All-atomistic modeling of highly crosslinked epoxy microstructures

Experimental data clarify that toughness enhancement by additives was related to the growth resistance of the localized cracks. To explore the molecular origin of crack resistance, in this section we adopt a computational simulation-based approach (see **Table S2** for a summary of the proposed multiscale framework). The neat epoxy and epoxy reinforced with toughening agents (epoxy-TA) models were fabricated in molecular dynamics simulation environment (Fig. 3(a)). The neat epoxy model consisted of 300 DGEBA resins and 150 EDA hardeners, with the same stoichiometry as the experiment. The epoxy-TA was composed of 300 DGEBA, 120 EDA, and 60C-8 diol toughening agents, which corresponds to a 20 mol% sample in the experiment (see **Supplementary Note 1** and **Fig. S6** for microstructure under transient condition, i.e., 30 and 40 mol% diol additive). Initially, all the molecular components were homogeneously distributed in the cubic unit cell. To avoid unintended intermolecular interferences during the initial positioning, the initial cell was prepared at an extremely low density of 0.2 g/cc. The unit cell was then gradually condensed over 5 ns using NPT ensemble at 300 K and 1 atm. A time step of 0.5 fs and the Verlet algorithm are used to describe the motion of the atoms.

For atomistic modeling and structural relaxation of the highly crosslinked epoxy network, the Polymer Consistent Force Field (PCFF) was employed [17]. PCFF, originally developed by Sun et al., is a class II type-based force field in which the four fundamental interactions governing polymer behavior—bond stretching, angle bending, dihedral torsion, and improper deformation—are explicitly parameterized using harmonic (or near-harmonic) potential functions. Under this framework, the polymer structure is described assuming small-amplitude fluctuations of covalent terms, while nonbonded interactions are treated using Lennard–Jones and Coulombic potentials. The harmonic approximation enables a stable and computationally efficient description of covalently bonded polymer structures with fixed bond topology. Owing to its extensive validation in predicting density, glass transition behavior, and thermal–mechanical properties of crosslinked thermoset systems [18], PCFF is particularly suitable for generating physically realistic and well-equilibrated network microstructures.

Subsequently, a crosslinking reaction was performed to form a three-dimensional network. The crosslinking reaction was performed using a distance-based algorithm [19–22]. The reactive sites were defined for each molecular component, and the distance between them is continuously monitored. When the relative distance is less than the given cutoff value, a new covalent bond is created at the corresponding reactive sites (Fig. 3(a) top). The cutoff starts at 3 Å and increases in 0.5 Å increments until 80% of all epoxide groups were reacted. For each increment, a short stabilization for 10 ps was performed in the NVT ensemble at 300 K

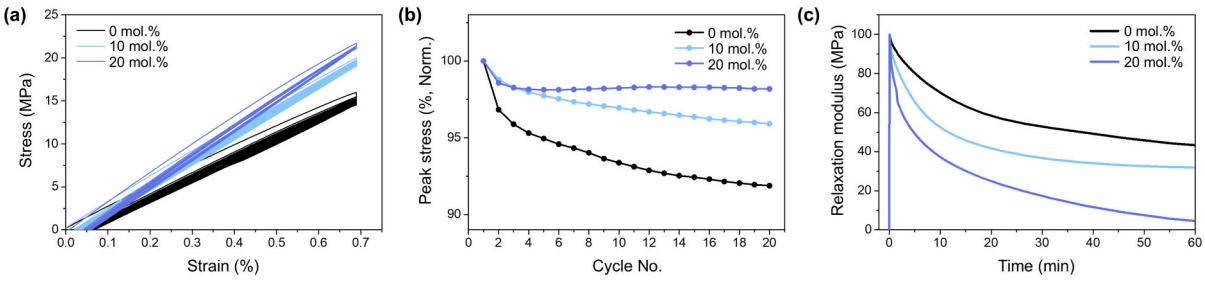


Fig. 2. Mechanical response and stress evolution under distinct loading conditions. (a) Stress–strain responses obtained during 20 loading-unloading cycles. (b) Cycle-dependent variation in the peak stress. (c) Stress relaxation behavior measured under a strain-holding condition.

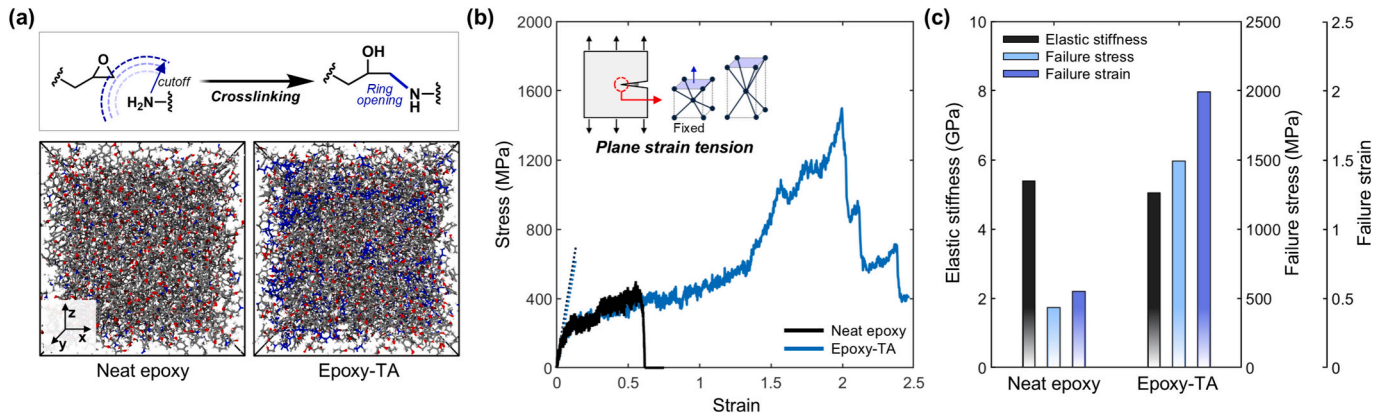


Fig. 3. Molecular dynamics-based investigation of toughening mechanisms. (a) Molecular configurations of epoxy microstructures reinforced with toughening agents. (b) Stress-strain response of neat epoxy and epoxy-TA under plane strain uniaxial deformation. The dotted line represents the slope in the elastic region. (c) Quantification of mechanical properties, including elastic stiffness, fracture stress, and failure strain.

and 1 atm. Lastly, the crosslinked neat epoxy and epoxy-TA models were further equilibrated at 300 K and 1 atm for 5 ns. The highly crosslinked structure was verified through the ratio of chains constituting the network. Molecular components are classified as crosslinked, dangling, and free chain according to their crosslinking state. In both neat epoxy and epoxy-TA, the ratios of each chain are in close agreement with previously reported ratios (**Table S3** [5,23–25]). Similarly, we further verified the proportion of amine groups (i.e., primary, secondary, tertiary amines) in the hardener molecule (**Table S4** [24–26]). The proportions of amine types were also in excellent agreement with previous measurements, indicating that the implemented model successfully reproduces the representative volume element of epoxy.

The experimental basis of this study clearly demonstrated that the epoxy systems form an amorphous and well-distributed network structure without aggregation or crystalline domain formation, even at additive contents as high as ~30 wt% [12]. The resulting specimens exhibited optical transparency and highly reproducible mechanical responses, indicating that microscopic compositional heterogeneity is minimal in the present organic system. Accordingly, the modeling framework adopted in this work assumes an ideally dispersed and homogeneous phase and does not explicitly incorporate stochastic modeling. Instead, the analysis focuses on elucidating the intrinsic mechanistic role of network topology modification under well-dispersed conditions, which represent the experimentally relevant and dominant structural state of the material.

Other simulation details are as follows: A commercial MD software, Materials Studio 2020 (Dassault Systèmes, France), was utilized to construct the thermoset microstructures. Subsequent mechanical tests were evaluated with the open-source MD code, LAMMPS (Sandia Labs, USA) [27]. Visualization of the microstructure was performed using the open visualization tool, OVITO [28]. The complete workflow of the work performed in the MD environment including atomic-level modeling and mechanical testing was provided in **Fig. S7** of Supplementary Data.

3.2. Mechanical responses of neat epoxy and topology-modified epoxy

The fracture response at the crack tip was analyzed under conditions of plane strain uniaxial deformation [29–31]. During the uniaxial elongation of the unit cell, the dimensions orthogonal to the loading direction were constrained to remain constant. The system temperature was regulated at 300 K using a Nosé–Hoover thermostat, and a constant engineering strain rate of 5×10^8 /s was applied (see **Supplementary Note 2** for consideration of strain rate). Throughout the deformation process, the atomic stress tensor was computed based on the virial theorem as follows [32]:

$$\sigma = \frac{1}{V}\left(-\sum_i^N m_i\left(v_i v_i^T\right) + \frac{1}{2}\sum_i^N \sum_{j\neq i}^N r_{ij} F_{ij} \right) \quad (1)$$

where V is the volume of system, N is the number of constituent atoms, m_i and v_i are the mass and velocity of i-th atom, and r_{ij} and F_{ij} indicate the relative position and interaction force between i-th and j-th atoms. The PCFF employed in the construction and stabilization process of epoxy models has inherent limitations in describing bond dissociation, as it models covalent bonds as permanently intact harmonic springs. Therefore, to accurately describe the fracture behavior of epoxy microstructures, we utilized a Reactive Force Field (ReaxFF) developed by van Duin et al [33,34] In contrast to classical force fields such as PCFF which assume fixed bond geometry, ReaxFF employs a bond-order formalism that allows for continuous bond breaking and formation during deformation. This capability allows for the capturing bond cleavage, crack initiation, and progressive failure mechanisms within polymer networks at the atomistic scale. Consequently, ReaxFF provides a realistic description of the fracture process, from chain damage in initial deformation to nanovoid formation and consolidation at high strain stages.

The mechanical response of neat epoxy and epoxy-TA under plane strain uniaxial deformation was evaluated independently (**Figs. 3(b)** and **3(c)**). The initial stress-strain behaviors of both materials displayed comparable trends. However, at higher strain levels ($\varepsilon > 0.5$), substantial divergence in their mechanical behavior was observed. Unlike the neat material, which exhibited premature failure after elastic-yielding, the epoxy thermoset modified with the toughening agent showed a distinct progression of elastic-yield-hardening-failure. Such stepwise changes of stress profiles are largely consistent with previously reported literatures [35–38].

3.3. Chain response analysis of hardeners and toughening agents

Above improvement in mechanical properties is associated with the distinct chain responses of the hardener and toughening agent. To investigate the microscopic responses in more detail, the end-to-end distance distributions of the hardener and the toughening agent were discretely evaluated (Fig. 4(a)). The distributions were normalized relative to the fully stretched molecular lengths of each component. In the neat epoxy model, the hardener molecules exhibit a bimodal distribution with two distinct peaks. This observation is attributed to the intrinsic chemical structure of the hardener (N-C-C-N), which adopts two predominant conformations: trans or gauche. Notably, the conformational flexibility of the hardener is inherently restricted, as the molecular architecture lacks sufficient degrees of freedom to form bent or

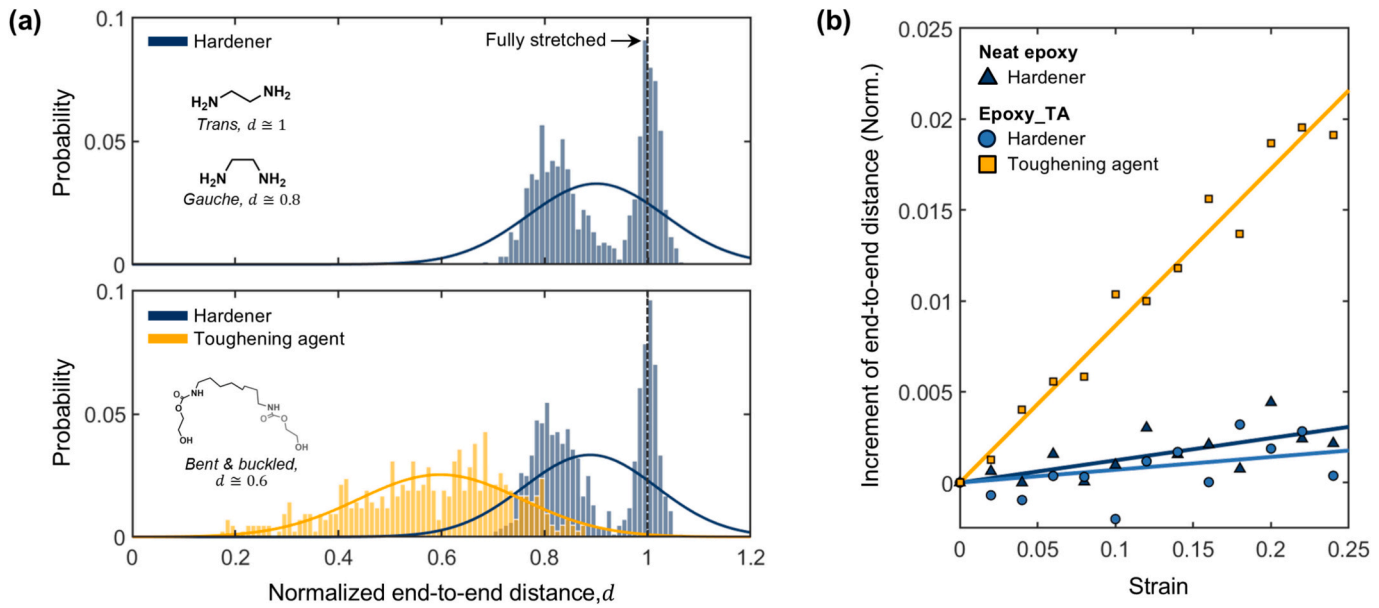


Fig. 4. Chain morphology analysis of the crosslinked epoxy microstructures. (a) End-to-end distance distributions of hardener and toughening agent in neat epoxy (top) and epoxy-TA (bottom). The solid line represents the Gaussian fit. (b) Increment of the average end-to-end distance during elongation. The solid line represents the linear fit.

buckled configurations. Consequently, the hardener molecules maintain an extended conformation within the polymer network (on average $d \cong 0.9$). In contrast, the toughening agent, which features a linear carbon backbone, possesses a greater degree of conformational freedom. This structural flexibility enables the formation of highly varied configurations, resulting in a lower average end-to-end distance (on average $d \cong 0.6$) and broader distribution deviations.

The external tensile forces applied to the molecular components are accommodated through two major mechanisms: the rearrangement of molecular configurations, primarily through chain extension, or intramolecular deformation. Chains that are already stretched in their equilibrium exhibit limited capacity for further extension under applied forces. For more clarity, the variations in the average end-to-end distances of the hardener and toughening agent were analyzed as functions of mechanical strain (Fig. 4(b)). The results reveal that the toughening agent with a flexible and buckled morphology demonstrates a pronounced increase in end-to-end distance with applied strain. Conversely, the hardener exhibits only a minor increase in the pre-extended distance. In other words, pre-extended hardener molecules experience greater intramolecular deformation, increasing the probability of bond cleavage and associated mechanical failure.

Asymmetric damage between the hardener and toughening agent was further investigated through strain energy density analysis. Since the virial stress tensor described in eq. (1) is essentially a per-atom quantity, the contributions can be individually subdivided for each molecular component in the system. The hardener and toughening agent components were independently grouped, and the stress tensor evolved in each molecule was derived. At this time, V in Eq. (1) is replaced by that of the corresponding component estimated from the Voronoi tessellation. The theoretical basis for such simulation analysis was provided in detail in our previous study [39,40]. During tensile deformation of the simulation cell, the strain energy density (u), which quantifies the substantial mechanical work of components, was defined as follows [41]:

$$u = \int_0^{\varepsilon_0} \sigma_{xx} d\varepsilon \tag{2}$$

where σ_{xx} indicates the virial stress component along the loading direction, and ε_0 is the uniaxial strain applied to the unit cell.

The results showed that the strain energy density was more pronounced in the pre-extended hardener component (Fig. 5(a)). Because of its inherent extension capacity, the toughening agent exhibits less than half the strain energy accumulation. For further characterization, the same analysis was performed on the backbone constituent atoms of each molecular component (Fig. 5(b)). In toughening agents, the strain energy density exhibited a decreasing trend toward the interior of the molecule. Specifically, substantial strain energy was observed from the crosslinking sites (O_1) to the urethane linkage with the triangular chemical structure (O_2-C_3-N); however, the transfer of strain energy was interrupted in the linear carbon segment (C_4-C_5-C_6-C_7) where chain flexibility is prominent. All backbone atoms of the hardener exhibit significant strain energy density, which is more pronounced at the crosslinking sites (N).

Furthermore, the reactive molecular dynamics allows continuous variation of covalent bond order during execution. In this regard, the magnitude of the structural distortion of the backbone during tension was quantified from the bond order parameter (Fig. 5(c)). The results are normalized to the order value at the equilibrium; thus, lower values indicate more severe deterioration. Overall, the bond order of constituent atoms was inversely proportional to the accumulated strain energy density. This was plausible because the forced elongation of covalent bonds due to the applied strain energy leads to a decrease in bond order. Notably, the atomic bond order within the toughening agent remained unaffected by mechanical loading. In other words, differences in conformational degrees of freedom lead to preferential bond cleavage in the hardener molecules (Fig. 5(d)).

3.4. Analysis of nanovoid distribution in epoxy networks during tension

Nano-sized free volumes formed at the fractured bond ends interrupt the load transfer of the epoxy thermoset, ultimately leading to mechanical failure. Therefore, the influence of asymmetric bond cleavage on nanovoid formation was analyzed. The nanovoid was estimated by the alpha-shape method with a probe sphere radius of 3.5 Å [42]. The nucleation and growth kinetics were evaluated by monitoring the number and fraction of free volume. During the initial loading stage, the total nanovoid volume fraction (V_{void}) remained negligible (Fig. 6(a) left). As the thermoset network undergoes elastic deformation, only

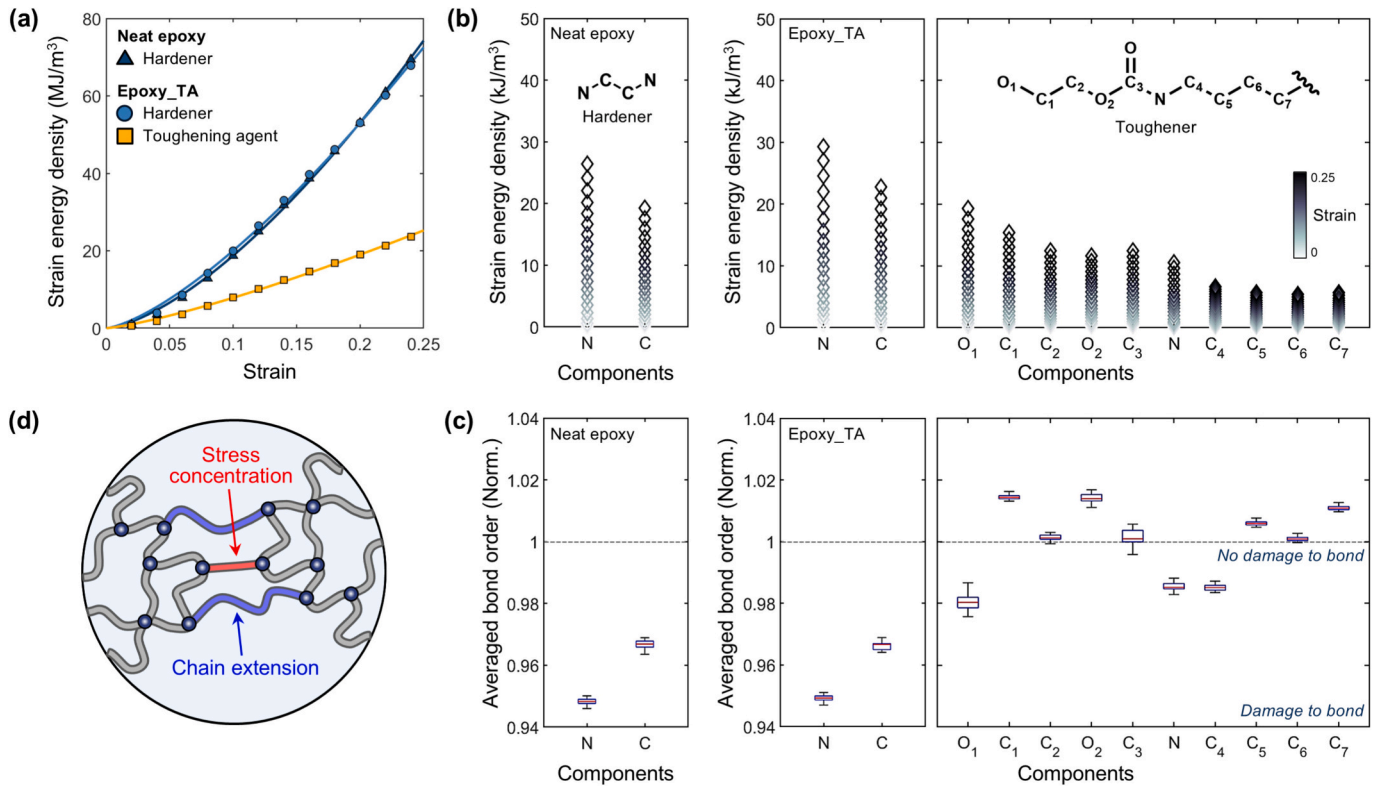


Fig. 5. Chain damage analysis of crosslinked epoxy microstructures. (a) Strain energy density evolved in the hardener and toughening agent during plane strain tension. The solid line is an exponential fitting of the results. (b) Accumulation of strain energy density of backbone constituent atoms. (c) Bond order distribution of backbone constituent atoms at mechanical strain, $\varepsilon = 0.25$. (d) Schematic for asymmetric structural deterioration between hardener and toughening agent.

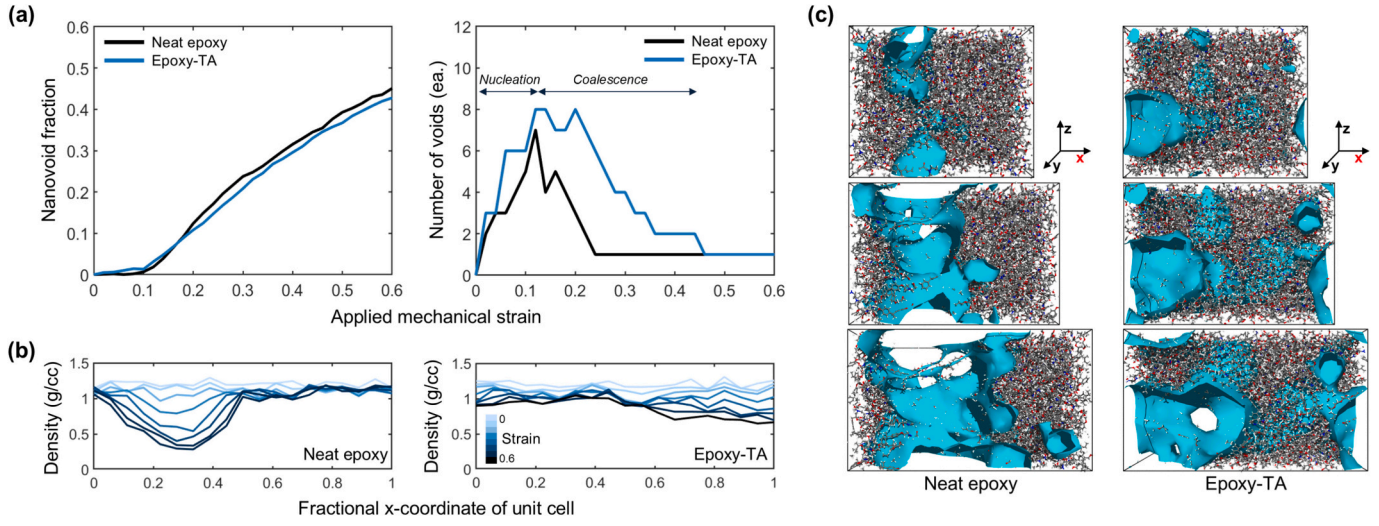


Fig. 6. Analysis of nanovoid formation in epoxy networks. (a) Changes in the number (left) and average size of nanovoids (right) during planar tension. (b) Spatial density changes of epoxy network along the loading axis (i.e., x-axis). (c) Snapshots of neat epoxy and epoxy-TA microstructures at mechanical strains of 0.2, 0.4, and 0.6. The free surfaces of voids are highlighted in cyan.

extremely fine nucleation sites are generated in the network (Fig. 6(a) right). However, upon exceeding a strain threshold of 0.1, a significant increase in nanovoid volume fraction was observed. The number of nanovoids also decreased significantly, indicating the rapid growth and coalescence of nanovoids due to the bond cleavage of the epoxy component.

Incorporation of a toughening agent notably influenced the nucleation and coalescence behavior of nanovoids. Specifically, the total nanovoid volume fractions of neat epoxy and epoxy-TA are very consistent under identical mechanical strains. However, epoxy-TA exhibits more active nucleation behavior compared to the neat materials. Due to the asymmetric bond damage, the toughening agent preserves the connectivity of the network even after cleavage of adjacent hardener molecules, preventing the opening of the fracture edges. This homogenizes the microscopic deformation of the network and effectively prevents the applied strain energy from being concentrated on a small number of fracture sites. The growth rate of individual nanovoids is effectively retards, thus the strain at which nanovoid coalescence is

completed is also effectively delayed (0.23 in neat epoxy and 0.46 in epoxy-TA).

The spatial distribution of nanovoids also shows clear differences (**Figs. 6(b)** and **6(c)**). In neat epoxy, nanovoids were distributed highly concentrated along planes perpendicular to the loading direction. Therefore, subsequent growth and coalescence of nanovoids bifurcated the thermoset network. This observation was supported by unit cell density distributions, which showed pronounced localized variations. Whereas epoxy-TA shows a homogeneous nanovoid distribution throughout the unit cell. Importantly, the integrated structure of nanovoid at high mechanical strain was also three-dimensional rather than planar. Such three-dimensional distribution of nanovoids preserved load transfer pathways within the matrix, further enhancing overall mechanical stability at large strain and fracture resistance. (see **Supplementary Note 3** and **Fig. S8** for details on the contributions of hardener and toughening agents to free volume formation).

3.5. Influence of topology modification in continuum-level mechanical fracture

To elucidate the role of topological modification in governing continuum-level mechanical failure, crack propagation simulations were carried out within a finite element framework. The linkage between subcontinuum and continuum descriptions was established by parameterizing specimen properties with inputs extracted from MD simulations. The most rigorous definition corresponds to the change in elastic energy associated with an incremental advance of the crack. Within an atomistic or molecular-scale framework, the elastic energy contributing to crack propagation can be decomposed into two primary components: (i) the intrinsic surface energy associated with the creation of new fracture surfaces, and (ii) the energy dissipated through structural evolution and irreversible deformation localized at the crack tip. For typical glassy polymers, it is well established that the energy required for craze initiation, fibril drawing, and subsequent fibril failure exceeds the intrinsic surface energy by orders of magnitude. Consequently, the work of fracture (G_c) is effectively governed by localized energy dissipation processes accompanying fibrillar deformation, stress transfer, and chain scission within the crack-tip craze.

In this context, the continuum mechanics model proposed by Brown, H. and Sha et al. provides a physically motivated framework to relate the macroscopic energy release to the micromechanical processes occurring within the craze [43,44]. By adopting a Dugdale zone approximation, the macroscopic energy release rate is equated to the work dissipated in the craze region at the crack tip. The craze is idealized as an elastic sheet, and the total energy required for crack propagation is obtained by integrating the strain energy stored and dissipated within this highly deformed region. To estimate the strain energy transmitted to an individual fibril at the crack tip, Brown's formulation incorporates two essential microstructural considerations. First, during plastic deformation, polymer chains reorganize into highly oriented load-bearing fibrils aligned with the tensile axis, while voids nucleate and are bridged by transverse secondary fibrils (crossties). Second, the resulting fibrillar network exhibits pronounced mechanical anisotropy, with the axial modulus along the fibril direction greatly exceeding the transverse modulus ($E_{11} \gg E_{22}$). Consistent with this framework, the network-modified system investigated in the present study exhibits early-stage chain scission and topological rearrangement under applied loading (Fig. 5), followed by the emergence of fibril-like domains preferentially aligned with the loading direction. The incorporation of the toughening agent promotes a more homogeneous distribution of nanovoids and the formation of distinct crosstie structures within the evolving network (Fig. 6). Since fibrillation is achieved at large deformation levels (strain ≈ 2; Fig. 3(b)), the constituent molecular chains experience substantial orientation and ordering. Moreover, the appearance of an extended stress plateau beyond the elastic regime provides additional evidence that fibrillation of the crosslinked network constitutes the dominant energy dissipation mechanism. Accordingly, the continuum-based model proposed by Brown offers an effective quantitative framework for capturing the fracture energy of the present network-modified system by consistently linking macroscopic energy release to microscale deformation and failure processes.

Specifically, the fracture energy was evaluated using the following equation:

$$G_c = 2\pi D \frac{{S_{max}}^2}{S}\left(1-\frac{1}{\lambda}\right)\sqrt{\frac{C_{11}}{C_{66}}} \tag{3}$$

where D denotes the characteristic spacing between adjacent fibrils, S_{max} and S correspond to the yield and maximum stresses, λ represents the average stretch ratio within the process zone, and C_{11} and C_{66} are the elastic tensile and in-plane shear moduli, respectively. The extraction of the parameters from molecular dynamics simulations followed the methodology reported in prior studies [30,31]. The fibril spacing D was quantified as the mean transverse distance between fibrils crossing the periodic boundary in the vicinity of fracture. The stress quantities S_{max} and S were obtained directly from the stress–strain response shown in Fig. 3(b). The stretch ratio λ was determined from the elongation at the strain level where the stress equals $(S+S_{max})/2$. To evaluate the in-plane shear modulus C_{66}, the deformation protocol described in Section 3.2 was repeated in the xy-plane, and the ratio $\sqrt{C_{11}/C_{66}}$ was subsequently computed as a function of strain (**Fig. S9**). The converged value of $\sqrt{C_{11}/C_{66}}$ was employed in Eq. (3) to incorporate the anisotropic mechanical response of the fibrillated process zone into the fracture energy calculation.

The parameters employed in the fracture energy evaluation are summarized in Table 1. The neat epoxy exhibited a fracture energy of 16.38 J/m^2, which was enhanced by nearly 20-fold upon incorporation of the toughening agent, reaching 321.81 J/m^2. The predicted values were highly comparable with the theorical and experimental range of conventional DGEBA-based epoxies (55–235 J/m^2 [45–49]), confirming the validity of the present approach. The enhancement in fracture energy was found to be strongly governed by the ratio ${S_{max}}^2/S$. In other words, the homogeneous nucleation and growth of nano-sized voids by the toughening agent maintain efficient load transfer at elevated strain levels, thereby significantly increasing the overall fracture resistance.

Finite element simulations of crack propagation were conducted on open-hole epoxy specimens (Fig. 7(a)). Employing fracture responses derived at the atomistic level into a damage model based on the traction-separation law has been proposed as a strategy for predicting continuum-level crack propagation [50–52]. To introduce stress concentration and confine the damage process within a well-defined region, a spherical hole of 30 μm radius was embedded at the specimen side. Poisson's ratio was considered to be 0.3, which is the experimental value for bulk epoxy [53,54]. Within Voigt notation, the isotropic elastic modulus was evaluated from the stiffness coefficient as $E = C_{11}(1+\nu)(1-2\nu)/(1-\nu)$, yielding 4.08 GPa. The lower boundary of the model was move-constrained, while a linear normal displacement was prescribed on the upper boundary to induce tensile loading. The only distinction between neat epoxy and epoxy-TA was the fracture energy parameter. Localized microstructural degradation was captured by a traction–separation law, activated once the maximum principal stress in a finite element exceeded 220 MPa, corresponding to the yield stress of both epoxies (Table 1). The fracture energy, expressed as the integral of the traction–separation profile, was calculated from the continuum mechanics model (see **Fig. S10** for constitutive law and damage model profiles).

Distinct crack propagation responses were observed between the neat epoxy and epoxy-TA specimens. In the early stages of deformation, both models exhibited purely elastic behavior, with pronounced stress concentration at the flanks of the open hole (Fig. 7(b)). Once the applied displacement exceeded approximately 8 μm, the local stress field

Table 1
Parameters of epoxy microstructure to calculate fracture energy.

	D (nm)	S (MPa)	S_{max} (MPa)	λ	$\sqrt{C_{11}/C_{66}}$	G_c (J/m^2)
Neat epoxy	4.45	223.6 ± 17.3	500.5 ± 20.6	1.361	1.97	16.38
Epoxy-TA	3.53	215.2 ± 23.1	1498.3 ± 2.7	2.079	2.68	321.81

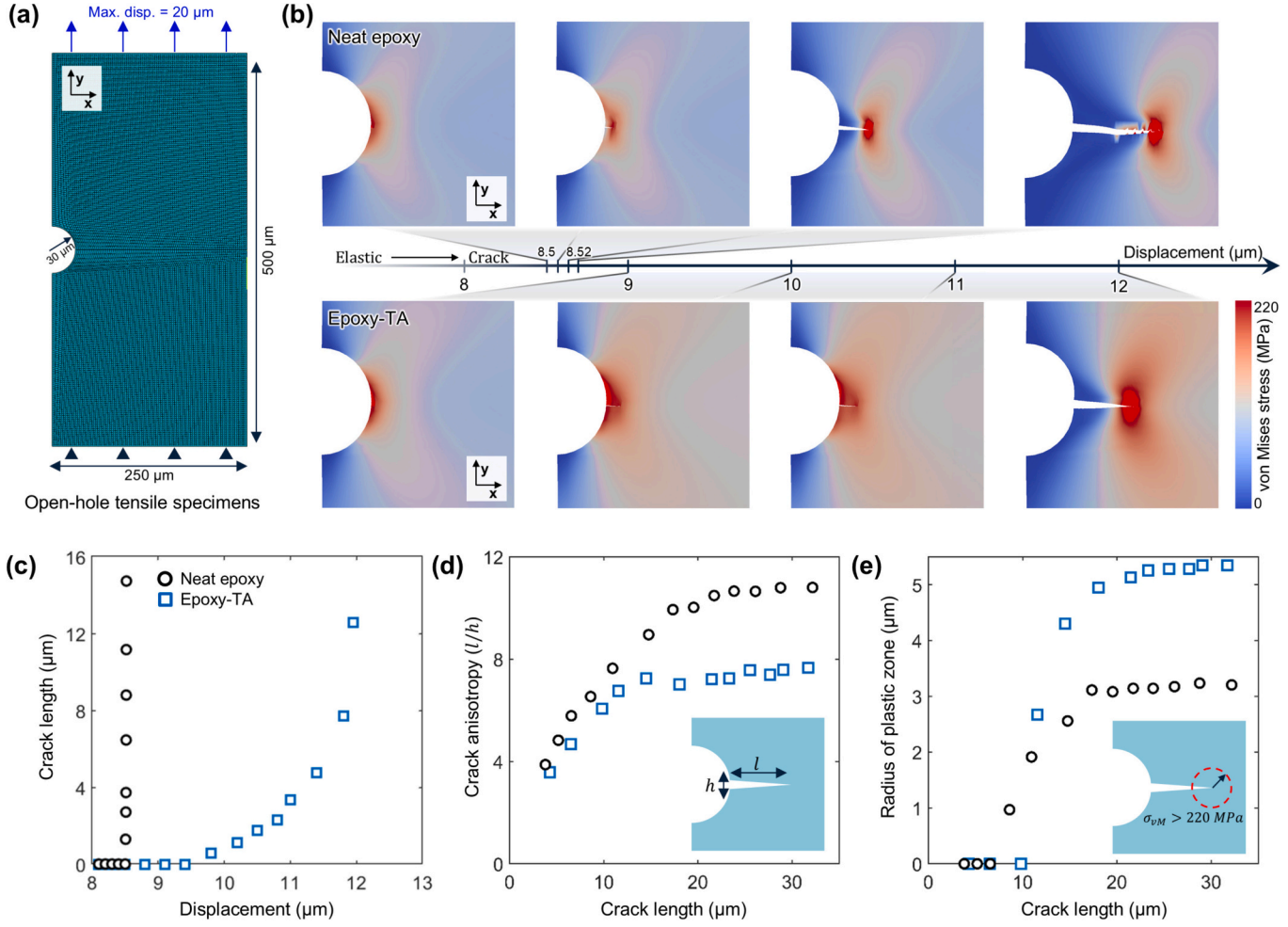


Fig. 7. Finite element simulation of crack propagation in neat epoxy and epoxy–TA systems. (a) Geometry and boundary conditions of the specimen model. (b) Representative snapshots illustrating crack progression under applied displacement. (c) Evolution of crack length as a function of displacement. (d) Variation of crack-tip anisotropy and plastic zone radius with crack extension.

surpassed the yield criterion, resulting in the initiation of damage at the hole boundary. The subsequent evolution of crack length revealed a pronounced difference in propagation kinetics between the two systems (Fig. 7(c)). For neat epoxy, crack extension occurred in a highly unstable manner, with rapid growth confined to a narrow displacement interval of 8.5–8.52 μm. By contrast, epoxy-TA accommodated additional loading, with crack progression distributed over a broader displacement range of 9–12 μm. This delayed and moderated crack advance highlighted the ability of the toughening agent to suppress catastrophic failure.

The origin of the observed toughness enhancement can be rationalized by two governing mechanisms of crack resistance: crack-tip anisotropy and the extent of the plastic zone. Crack-tip anisotropy was quantified as the ratio of the longitudinal to the transverse crack length (l/h) (Fig. 7(d)). This parameter increased progressively with crack extension and converges as the system transitions into a steady growth regime. In neat epoxy, the anisotropy ratio exhibited a pronounced and rapid escalation, indicative of preferential elongation of the crack front along the loading transverse axis. This in turn exacerbated stress localization and promoted unstable crack advance. In contrast, epoxy–TA displayed a more gradual evolution of anisotropy, suggesting a redistribution of strain fields in the crack-tip that effectively mitigates stress concentration.

The plastic zone, the region where the crosslinked network undergoes fibrillation, served as a primary source of energy dissipation during deformation. The spatial extent of plastic zone was estimated by calculating the effective radius of the region exceeding the yield stress threshold of 220 MPa (Fig. 7(e)). Both systems exhibited a progressive enlargement of the plastic zone followed by convergence at steady crack growth. Crucially, the plastic zone radius in epoxy-TA was approximately 1.7 times greater than that in neat epoxy model, indicating its superior capacity for mechanical energy dissipation. This was also consistent with experimental observations showing a rough fracture surface with the content of toughening agent (Fig. 1(d)). Collectively, the suppression of crack-tip anisotropy and the enlargement of the plastic zone act synergistically to inhibit catastrophic crack propagation

and enhance fracture resistance in epoxy-TA.

4. Conclusion

In this study, we investigated how topology modification via homogeneous additives governs the fracture resistance of epoxy thermosets. The results demonstrate that toughness enhancement does not arise from uniformly strengthening the network, but from deliberately introducing damage asymmetry that regulates where and how failure initiates and evolves. By explicitly coupling atomistic bond scission with continuum-scale fracture energetics, the present multiscale framework reveals that preferential damage localization within the hardener—while preserving the structural integrity of the flexible additive—fundamentally redirects the fracture pathway of the network.

Beyond mechanistic clarification, this work establishes topology modification as a quantitative design variable for controlling failure modes in crosslinked polymers. Directly linking molecular-scale bond dynamics to macroscopic fracture metrics enables predictive optimization of network architectures prior to synthesis, thereby moving beyond phenomenological toughening strategies. Consequently, the findings provide actionable design principles and a computational framework for developing next-generation structural polymers with tailored fracture resistance.

The current study provides topology design guidance for enhancing toughness, focusing on quasi-static conditions. However, practical applications of network polymers often involve dynamic and impact loading, where rate-dependent effects become critical [55–57]. Future work should therefore address the role of topology and toughening agents under non-equilibrium conditions, with particular attention to strain-rate–induced heterogeneity and thermo-mechanical coupling at the molecular level.

CRediT authorship contribution statement

Hongdeok Kim: Writing – original draft, Visualization, Software, Methodology, Data curation. **Gyuri Kim:** Writing – original draft, Validation, Formal analysis, Data curation. **Min Sang Kwon:** Writing – review & editing, Supervision, Project administration. **Joonmyung Choi:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that may have influenced the work reported in this study.

Acknowledgments

Hongdeok Kim and Gyuri Kim contributed equally to this work. This study was supported by the National Research Foundation of Korea (NRF) grant (RS-2023-00210865) funded by the Korean government (MSIT). This research was supported by Korea Basic Science Institute (National research Facilities and Equipment Center) grant funded by the Korea Ministry of Education (Grant No. RS-2025-02314069).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.173719.

Data availability

Data will be made available on request.

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